

# Christopher A. Choquette-Choo

 [christopherchoquette.com](https://christopherchoquette.com)  
 [christopher-choquette-choo](https://in.linkedin.com/in/christopher-choquette-choo)

 [cchoquette](https://github.com/cchoquette)  
 CA, USA

**Research Scientist** 50+ papers, lead product deployments.

Significant contributions to 10+ major products with billions of users and enabling 100s of downstream usecases. I am a research scientist and engineer. I enjoy defining then solving tough problems, and deploying the solutions.

## Research Experience

---

### OpenAI

Research Scientist (Member of Technical Staff) on the Alignment team

San Francisco, CA, USA

2025 – Present

- Led GPT-Red. Training, evaluation, release, etc. I was the main writer of [this](#).
- Beyond, I lead our automated red teaming and robustness research effort.
- I focus particularly on RL, evals, grading, priors, and other related things.
- Broadly, I work on alignment of frontier LLMs, including safety, security, privacy, and model/system robustness. infrastructure design, evaluations, data, etc.
- I have had significant contributions to GPT-5.2, 5.3, 5.4, 5.5, 5.6, ... I led several robustness contributions.

### Google Brain & Google DeepMind

Senior Research Scientist

Research Scientist

Machine Learning Researcher

Mountain View, CA, USA

2024 – 2025

2024 – 2024

2022 – 2023

- Lead privacy audits for frontier models. Grew this from Google DeepMind to across all of Google. Directly enable product releases for 100s of products through rigorous compliance testing.
- Lead security efforts, in particular, for security of agents, e.g., project Mariner. Designed and implemented mitigations, attacks, and benchmarks.
- Contribute to frontier models via data, training algorithms, and evaluations, e.g., Gemini, Gemma, GBoard, PaLM, etc. A focus on better privacy and security.
- Research memorization, privacy/security vulnerabilities, and auditing of ML/language models.
- Research and develop state-of-the-art differential privacy mechanisms for machine learning.
- Lead research into compression in federated learning.
- Deploy my techniques for compression, memorization analysis, and differential privacy into production.
- 11 spot bonuses for exceptional work, including LLM releases, impactful research like DP-FTRL, and attacking SOTA models like GPT-3, and leading privacy evals outside GDM.
- 1000+ CLs, 1 competition, 50+ papers released to date, 3 patents.

### Google Research, Cerebra team

Brain Resident

New York, NY, USA

2020 – 2022

- Investigated concept interpretability of acoustic models. Presented at Google Research Conference.
- Led research into optimal privacy-communication-accuracy tradeoffs with sparsity in federated learning.
- Researched differentially private multi-winner voting mechanisms for machine learning.
- Guided and advise project into private semi-supervised learning for federated learning in dermatology.

### Vector Institute, with Professor Nicolas Papernot

Research Assistant

Toronto, ON, Canada

Sept 2019 – Oct. 2020

- Led research into differentially private collaborative algorithms.
- Led Privacy-preserving machine learning.

### Georgian Partners

Research Engineer

Toronto, ON, Canada

Apr. 2019 – Aug. 2019

- Owned development of a differentially private ML model, to guarantee user data privacy, in collaboration with Google's top machine learning library, TensorFlow/Privacy, which is used by 1000 people.
- Designed an AutoML package to intelligently tune an ML model on any dataset; used by 25+ people.

**Vector Institute**, with Professor Aspuru-Guzik  
Undergraduate Researcher

Toronto, ON, Canada  
Apr. 2019 – Aug. 2019

- Researched machine learning for molecular discovery via Gaussian processes and active learning.

**Intel Corp.**  
Research Engineer

Toronto, ON, Canada  
May 2018 – May 2019

- Spearheaded SOTA ML bug triager with 55% accuracy on 2000+ engineers and 76% on 500+ teams.
- Productionized triager with an engineering efficiency improvement of 25% and savings of >\$10M annually.

**Institute of Biomaterials and Biomedical Engineering** with Professor Paul Santerre  
Undergraduate Researcher

Toronto, ON, Canada  
Apr. 2016 – Sept. 2016

- Studied mechanical properties of polyurethane scaffolds and dental resin composites. Used in patents.

## Research and Papers

[X] = First or Co-First Author. To date, I've first or co-first authored 14 papers.

### Peer-Reviewed Conference and Journal Proceedings

[61] *Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy, Research, and Practice* [Link](#) 2025  
Thirty-ninth Conference on Neural Information Processing Systems (Neurips)

A. Feder Cooper\*, **Christopher A. Choquette-Choo**\*, Miranda Bogen\*, Matthew Jagielski\*, Katja Filippova\*, Ken Ziyu Liu\*, ..., Nicolas Papernot, Katherine Lee

\*Equal contribution.

[60] *Strong Membership Inference Attacks on Massive Datasets and (Moderately) Large Language Models* [Link](#) 2025  
Thirty-ninth Conference on Neural Information Processing Systems (Neurips)

Jamie Hayes, Ilia Shumailov, **Christopher A. Choquette-Choo**, Matthew Jagielski, George Kaissis, Katherine Lee, Milad Nasr, Sahra Ghalebikesabi, Niloofar Mireshghallah, Meenatchi Sundaram Mutu Selva Annamalai, Igor Shilov, Matthieu Meeus, Yves-Alexandre de Montjoye, Franziska Boenisch, Adam Dziedzic, A. Feder Cooper

[59] *Language Models May Verbatim Complete Text They Were Not Explicitly Trained On* [Link](#) 2025  
(Spotlight) Proceedings of the 42nd International Conference on Machine Learning (ICML)

Ken Liu, **Christopher A. Choquette-Choo**\*, Matthew Jagielski\*, Peter Kairouz, Sanmi Koyejo, Nicolas Papernot, Percy Liang

\*Equal contribution.

[58] *Scaling Laws for Differentially Private Language Models* [Link](#) 2025  
Proceedings of the 42nd International Conference on Machine Learning (ICML)

Ryan McKenna, Yangsibo Huang, Amer Sinha, Borja Balle, Zachary Charles, **Christopher A. Choquette-Choo**, Badih Ghazi, Georgios Kaissis, Ravi Kumar, Ruibo Liu, Da Yu, Chiyuan Zhang

[57] *Correlated Noise Mechanisms for Differentially Private Learning* [Link](#) 2026  
Foundations and Trends in Machine Learning

Krishna Pillutla, Jalaj Upadhyay, **Christopher A. Choquette-Choo**, Krishnamurthy Dvijotham, Arun Ganesh, Monika Henzinger, Jonathan Katz, Ryan McKenna, H. Brendan McMahan, Keith Rush, Thomas Steinke, Abhradeep Thakurta

[56] *Exploring and Mitigating Adversarial Manipulation of Voting-Based Leaderboards* [Link](#) 2025  
(Oral) Proceedings of the 42nd International Conference on Machine Learning (ICML)

Yangsibo Huang, Milad Nasr, Anastasios Angelopoulos, Nicholas Carlini, Wei-Lin Chiang, **Christopher A. Choquette-Choo**, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Ken Ziyu Liu, Ion Stoica, Florian Tramèr, Chiyuan Zhang

- [55] *POST: A Framework for Privacy of Soft-prompt Transfer* [Link](#) 2025  
Proceedings of the 42nd International Conference on Machine Learning (ICML)  
Xun Wang, Jing Xu, **Christopher A. Choquette-Choo**, Adam Dziedzic, Franziska, Boenisch
- [54] *Privacy Ripple Effects from Adding or Removing Personal Information in Language Model Training* [Link](#) 2025  
The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)  
Jaydeep Borkar, Katherine lee, Matthew Jagielski, David A. Smith, **Christopher A. Choquette-Choo**
- [53] *Measuring memorization in language models via probabilistic extraction* [Link](#) 2025  
2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics  
Jamie Hayes, Marika Swanberg, Harsh Chaudhari, Itay Yona, Ilia Shumailov, Milad Nasr, **Christopher A. Choquette-Choo**, Katherine Lee, A. Feder Cooper
- [52] *Scalable Extraction of Training Data from (Production) Language Models* [Link](#) 2025  
The Thirteenth International Conference on Learning Representations (ICLR)  
Milad Nasr, Nicholas Carlini, Jonathan Hayase, Matthew Jagielski, A. Feder Cooper, Daphne Ippolito, **Christopher A. Choquette-Choo**, Eric Wallace, Florian Tramèr, Katherine Lee
- [51] *Privacy Auditing of Large Language Models* [Link](#) 2025  
The Thirteenth International Conference on Learning Representations (ICLR)  
Ashwinee Panda, Xinyu Tang, Milad Nasr, **Christopher A. Choquette-Choo**, Prateek Mittal
- [50] *Near Exact Privacy Amplification for Matrix Mechanisms* [Link](#) 2025  
The Thirteenth International Conference on Learning Representations (ICLR)  
**Christopher A. Choquette-Choo**, Arun Ganesh, Saminul Haque, Thomas Steinke, Abhradeep Thakurta
- [49] *Optimal Rates for DP-SCO with a Single Epoch and Large Batches* [Link](#) 2025  
The 36th International Conference on Algorithmic Learning Theory (ALT)  
**Christopher A. Choquette-Choo**, Arun Ganesh, Abhradeep Thakurta
- [48] *Recite, Reconstruct, Recollect: Memorization in LMs as a Multifaceted Phenomenon* [Link](#) 2025  
The Thirteenth International Conference on Learning Representations (ICLR)  
USVSN Sai Prashanth, Alvin Deng, Kyle O'Brien, Jyothir S V, Mohammad Aflah Khan, Jaydeep Borkar, **Christopher A. Choquette-Choo**, Jacob Ray Fuehne, Stella Biderman, Tracy Ke, Katherine Lee, Naomi Saphra
- [47] *The Last Iterate Advantage: Empirical Auditing and Principled Heuristic Analysis of Differentially Private SGD* [Link](#) 2025  
The Thirteenth International Conference on Learning Representations (ICLR)  
Milad Nasr, Thomas Steinke, Borja Balle, **Christopher A. Choquette-Choo**, Arun Ganesh, Matthew Jagielski, Jamie Hayes, Abhradeep Thakurta, Adam Smith, Andreas Terzis
- [46] *User Inference Attacks on Large Language Models* [Link](#) 2024  
The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)  
Nikhil Kandpal, Krishna Pillutla, Alina Oprea, Peter Kairouz, **Christopher A. Choquette-Choo**, Zheng Xu
- [45] *Auditing Private Prediction* [Link](#) 2024  
Proceedings of the 41st International Conference on Machine Learning (ICML)  
Karan Chadha, Matthew Jagielski, Nicolas Papernot, **Christopher A. Choquette-Choo**, Milad Nasr
- [44] *Privacy Side-Channels in Machine Learning Systems* [Link](#) 2024  
USENIX Security Symposium (USENIX)  
Edoardo Debenedetti, Giorgio Severi, Milad Nasr, **Christopher A. Choquette-Choo**, Matthew Jagielski, Eric Wallace, Nicholas Carlini, Florian Tramèr

- [43] *Privacy Amplification for Matrix Mechanisms* [Link](#) 2024  
(Spotlight) International Conference on Learning Representations (ICLR)  
**Christopher A. Choquette-Choo**, Arun Ganesh, Thomas Steinke, Abhradeep Guha Thakurta
- [42] *Correlated Noise Provably Beats Independent Noise for Differentially Private Learning* [Link](#) 2024  
International Conference on Learning Representations (ICLR)  
**Christopher A. Choquette-Choo**, Krishnamurthy Dj Dvijotham, Krishna Pillutla, Arun Ganesh, Thomas Steinke, Abhradeep Guha Thakurta
- [41] *Teach LLMs to Phish: Stealing Private Information from Language Models* [Link](#) 2024  
International Conference on Learning Representations (ICLR)  
Ashwinee Panda, **Christopher A. Choquette-Choo**, Zhengming Zhang, Yaoqing Yang, Prateek Mittal
- [40] *Poisoning web-scale training datasets is practical* [Link](#) 2024  
IEEE Symposium on Security and Privacy (IEEE S&P)  
Nicholas Carlini, Matthew Jagielski, **Christopher A. Choquette-Choo**, Daniel Paleka, Will Pearce, Hyrum Anderson, Andreas Terzis, Kurt Thomas, Florian Tramèr.
- [39] *(Amplified) Banded Matrix Factorization: A unified approach to private training* [Link](#) 2023  
Thirty-seventh Conference on Neural Information Processing Systems (Neurips)  
**Christopher A. Choquette-Choo**, Arun Ganesh, Ryan McKenna, H. Brendan McMahan, Keith Rush, Abhradeep Guha Thakurta, Zheng Xu.
- [38] *Are aligned neural networks adversarially aligned?* [Link](#) 2023  
Thirty-seventh Conference on Neural Information Processing Systems (Neurips)  
Nicholas Carlini, Milad Nasr, **Christopher A. Choquette-Choo**, Matthew Jagielski, Irena Gao, Anas Awadalla, Pang Wei Koh, Daphne Ippolito, Katherine Lee, Florian Tramèr, Ludwig Schmidt.
- [37] *Students Parrot Their Teachers: Membership Inference on Model Distillation* [Link](#) 2023  
(Oral) Thirty-seventh Conference on Neural Information Processing Systems (Neurips)  
Matthew Jagielski, Milad Nasr, Katherine Lee, **Christopher A. Choquette-Choo**, Nicholas Carlini.
- [36] *MADLAD-400: Multilingual And Document-Level Large Audited Dataset* [Link](#) 2023  
Thirty-seventh Conference on Neural Information Processing Systems (Neurips)  
Sneha Kudugunta, Isaac Caswell, Biao Zhang, Xavier Garcia, **Christopher A. Choquette-Choo**, Katherine Lee, Derrick Xin, Aditya Kusupati, Romi Stella, Ankur Bapna, Orhan Firat
- [35] *Robust and Actively Secure Serverless Collaborative Learning* [Link](#) 2023  
Thirty-seventh Conference on Neural Information Processing Systems (Neurips)  
Nicholas Franzese, Adam Dziedzic, **Christopher A. Choquette-Choo**, Mark R. Thomas, Muhammad Ahmad Kaleem, Stephan Rabanser, Congyu Fang, Somesh Jha, Nicolas Papernot, Xiao Wang
- [34] *Multi-epoch matrix factorization mechanisms for private machine learning* [Link](#) 2023  
(Oral) Proceedings of the 40th International Conference on Machine Learning (ICML)  
**Christopher A. Choquette-Choo**, H. Brendan McMahan, Keith Rush, Abhradeep Thakurta.
- [33] *Private Federated Learning with Autotuned Compression* [Link](#) 2023  
Proceedings of the 40th International Conference on Machine Learning (ICML)  
Enayat Ullah\*, **Christopher A. Choquette-Choo**\*, Peter Kairouz\*, Sewoong Oh\*.  
\*Equal contribution
- [32] *Federated Learning of Gboard Language Models with Differential Privacy* [Link](#) 2023  
The 61st Annual Meeting of the Association for Computational Linguistics  
Zheng Xu, Yanxiang Zhang, Galen Andrew, **Christopher A. Choquette-Choo**, Peter Kairouz, H. Brendan McMahan, Jesse Rosenstock, Yuanbo Zhang.
- [31] *Preventing verbatim memorization in language models gives a false sense of privacy* [Link](#) 2023  
(Runner-up Best Paper) Proceedings of the 16th International Natural Language Generation Conference

Daphne Ippolito, Florian Tramèr\*, Milad Nasr\*, Chiyuan Zhang\*, Matthew Jagielski\*, Katherine Lee\*,  
**Christopher A. Choquette-Choo\***, Nicholas Carlini.

\*Equal contribution, random ordering.

[30] *Proof-of-Learning is Currently More Broken Than You Think* [Link](#) 2023

IEEE 8th European Symposium on Security and Privacy (EuroS&P). IEEE Computer Society

Congyu Fang\*, Hengrui Jia\*, Anvith Thudi, Mohammad Yaghini, **Christopher A. Choquette-Choo**,  
Natalie Dullerud, Varun Chandrasekaran, Nicolas Papernot.

\*Equal contribution, alphabetical ordering.

[29] *Private Multi-Winner Voting for Machine Learning* [Link](#) 2023

Proceedings on 23rd Privacy Enhancing Technologies Symposium (PETS)

Adam Dziedzic, **Christopher A. Choquette-Choo**, Natalie Dullerud, Vinith Menon Suriyakumar, Ali  
Shahin Shamsabadi, Muhammad Ahmad Kaleem, Somesh Jha.

[28] *The fundamental price of secure aggregation in differentially private federated learning* [Link](#) 2022

(Spotlight) International Conference on Machine Learning. PMLR

Wei-ning Chen\*, **Christopher A. Choquette-Choo\***, Peter Kairouz\*, Ananda Theertha Suresh\*.

\*Equal contribution, alphabetical ordering.

[27] *Label-Only Membership Inference Attacks* [Link](#) 2021

(Spotlight) International Conference on Machine Learning (ICML)

**Christopher A. Choquette-Choo**, Florian Tramer, Nicholas Carlini, Nicolas Papernot.

[26] *Entangled Watermarks as a Defense against Model Extraction* [Link](#) 2021

USENIX Security Symposium (USENIX)

Hengrui Jia, **Christopher A. Choquette-Choo**, Varun Chandrasekaran, Nicolas Papernot.

[25] *Proof of Learning: Definitions and Practice* [Link](#) 2021

IEEE Symposium on Security and Privacy (IEEE S&P)

Hengrui Jia\*, Mohammad Yaghini\*, **Christopher A Choquette-Choo**, Natalie Dullerud, Anvith Thudi,  
Varun Chandrasekaran, Nicolas Papernot.

\*Equal contribution, alphabetical ordering.

[24] *Machine Unlearning* [Link](#) 2021

IEEE Symposium on Security and Privacy (IEEE S&P)

Lucas Bourtole\*, Varun Chandrasekaran\*, **Christopher A. Choquette-Choo\***, Hengrui Jia\*, Adelin  
Travers\*, Baiwu Zhang\*, David Lie, Nicolas Papernot.

\*Equal contribution, alphabetical ordering.

[23] *CaPC Learning: Confidential and Private Collaborative Learning* [Link](#) 2021

International Conference on Learning Representations (ICLR)

**Christopher A. Choquette-Choo\***, Natalie Dullerud\*, Adam Dziedzic\*, Yunxiang Zhang\*, Somesh  
Jha, Nicolas Papernot, Xiao Wang.

\*Equal contribution, alphabetical ordering.

[22] *A Multi-label, Dual-Output Deep Neural Network for Automated Bug Triaging* [Link](#) 2019

International Conference on Machine Learning and Applications (ICMLA)

**Christopher A. Choquette-Choo**, David Sheldon, Jonny Proppe, John Alphonso-Gibbs, Harsha  
Gupta.

### **Peer-Reviewed Workshop Proceedings**

[21] *Privacy Auditing of Large Language Models* [Link](#) 2024

Next Generation of AI Safety Workshop at ICML 2024

Ashwinee Panda, Xinyu Tang, Milad Nasr, **Christopher A. Choquette-Choo**, Prateek Mittal

[20] *Privacy Auditing of Large Language Models* [Link](#) 2024

FM-Wild Workshop at ICML 2024

- [19] *User Inference Attacks on Large Language Models* [Link](#) 2023  
International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with NeurIPS (FL@FM-NeurIPS'23)  
Nikhil Kandpal, Krishna Pillutla, Alina Oprea, Peter Kairouz, **Christopher A. Choquette-Choo**, Zheng Xu
- [18] *Correlated Noise Provably Beats Independent Noise for Differentially Private Learning* [Link](#) 2023  
International Workshop on Federated Learning in the Age of Foundation Models (FL@FM-NeurIPS'23)  
**Christopher A. Choquette-Choo**, Krishnamurthy Dj Dvijotham, Krishna Pillutla, Arun Ganesh, Thomas Steinke, Abhradeep Guha Thakurta
- [17] *User Inference Attacks on Large Language Models* [Link](#) 2023  
Socially Responsible Language Modelling Research (SoLaR)  
Nikhil Kandpal, Krishna Pillutla, Alina Oprea, Peter Kairouz, **Christopher A. Choquette-Choo**, Zheng Xu
- [16] *Communication Efficient Federated Learning with Secure Aggregation and Differential Privacy* [Link](#) 2021  
the Neural Information Processing Systems (NeurIPS) workshop on Privacy in Machine Learning  
Wei-ning Chen\*, **Christopher A. Choquette-Choo**\*, Peter Kairouz\*.  
\*Equal contribution, alphabetical ordering.

## Reports

- [15] *VaultGemma: A Differentially Private Gemma Model* [Link](#) 2025  
arxiv  
..., **Christopher A. Choquette-Choo**\*, ...  
\*Contributor. Led memorization and agentic security efforts.
- [14] *Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities.* [Link](#) 2025  
arxiv  
..., **Christopher A. Choquette-Choo**\*, ...  
\*Contributor. Led memorization and agentic security efforts.
- [13] *Lessons from Defending Gemini Against Indirect Prompt Injections* [Link](#) 2025  
arXiv  
Chongyang Shi, Sharon Lin, Shuang Song, Jamie Hayes, Ilia Shumailov, Itay Yona, Juliette Pluto, Aneesh Pappu, **Christopher A. Choquette-Choo**, Milad Nasr, Chawin Sitawarin, Gena Gibson, Andreas Terzis, John "Four" Flynn
- [12] *Gemma 3 Technical Report* [Link](#) 2025  
arxiv  
..., **Christopher A. Choquette-Choo**\*, ...  
\*Contributor. Led memorization efforts.
- [11] *Gemma 2: Improving Open Language Models at a Practical Size* [Link](#) 2024  
arxiv  
..., **Christopher A. Choquette-Choo**\*, ...  
\*Contributor. Led memorization efforts.
- [10] *CodeGemma: Open Code Models Based on Gemma* [Link](#) 2024  
arXiv  
..., **Christopher A. Choquette-Choo**\*, ...  
\*Contributor.
- [9] *Gemma: Open Models Based on Gemini Research and Technology* [Link](#) 2024  
arXiv



..., **Christopher A. Choquette-Choo\***, ...

\*Contributor. Led memorization efforts.

[8] *Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context* [Link](#) 2024  
arXiv

..., **Christopher A. Choquette-Choo\***, ...

\*Contributor. Led memorization testing.

[7] *Gemini: A Family of Highly Capable Multimodal Models* [Link](#) 2023  
arXiv

Anil, R., ..., **Christopher A. Choquette-Choo\***, ..., & Vinyals, O.

\*Contributor. Led memorization efforts.

[6] *Palm 2 technical report* [Link](#) 2023  
arXiv

Anil, R., Dai, A. M., Firat, O., Johnson, M., Lepikhin, D., Passos, A., ..., **Christopher A. Choquette-Choo\***, ..., & Wu, Y.

\*Core contributor. Led memorization efforts.

[5] *Report of the 1st Workshop on Generative AI and Law* [Link](#) 2023  
arXiv

A. Feder Cooper\*, Katherine Lee\*, James Grimmelmann, Daphne Ippolito, Christopher Callison-Burch, **Christopher A. Choquette-Choo**, ...

\*Equal contribution, alphabetical ordering.

### **Pre-Prints (arXiv)**

[4] *Extracting alignment data in open models* [Link](#) 2025  
arXiv

Federico Barbero, Xiangming Gu, **Christopher A. Choquette-Choo**, Chawin Sitawarin, Matthew Jagielski, Itay Yona, Petar Veličković, Ilia Shumailov, Jamie Hayes

[3] *LLMs unlock new paths to monetizing exploits* [Link](#) 2025  
arXiv

Nicholas Carlini, Milad Nasr, Edoardo Debenedetti, Barry Wang, **Christopher A. Choquette-Choo**, Daphne Ippolito, Florian Tramèr, Matthew Jagielski

[2] *Phantom: General Trigger Attacks on Retrieval Augmented Language Generation* [Link](#) 2024  
arXiv

Harsh Chaudhari, Giorgio Severi, John Abascal, Matthew Jagielski, **Christopher A. Choquette-Choo**, Milad Nasr, Cristina Nita-Rotaru, Alina Oprea

[1] *Fine-tuning with differential privacy necessitates an additional hyperparameter search* [Link](#) 2022  
arXiv

Yannis Cattan, **Christopher A Choquette-Choo**, Nicolas Papernot, Abhradeep Thakurta

### **Under Review (and not yet released)**

## **Talks**

---

### **Invited Talks**

**DP-Follow-The-Regularized-Leader: State-of-the-art Optimizers for Private Machine Learning.** 2024  
*Institute of Science and Technology Austria (ISTA) for Prof. Christoph Lampert* Slides available upon request.

**DP-Follow-The-Regularized-Leader: State-of-the-art Optimizers for Private Machine Learning.** 2024  
"Federated Learning on the Edge" AAAI Spring 2024 Symposium. Slides available upon request.

**Host of "Private Optimization with Correlated Noise" invited session and co-presented first talk** 2024  
Information Theory and Applications (ITA) Slides available upon request.

**Poisoning Web-Scale Training Datasets is Practical** 2024  
Guest talk for Prof. Varun Chandrasekaran at University of Illinois Slides available upon request.

**The Privacy Considerations of Production Machine Learning** 2021  
MLOps New York Area Summit Slides available upon request.

**Adversarial Machine Learning: Ensuring Security and Privacy of ML Models and Sensitive Data** 2019  
REWORK Responsible AI Summit [Available as a part of the Privacy and Security in Machine Learning package](#)

### **Paper Presentations**

Multi-Epoch Matrix Factorization Mechanisms for Private Machine Learning [Oral presentation at ICML 2023](#)  
The Fundamental Price of Secure Aggregation in Differentially Private Machine Learning [Spotlight at ICML 2022](#)  
Label-Only Membership Inference Attacks [Spotlight at ICML 2021](#)  
Proof-of-Learning Definitions and Practice [Oral presentation at IEEE S&P 2021](#)  
Machine Unlearning [Oral presentation at IEEE S&P 2021](#)

### **Professional Activities**

---

#### **Program Committee**

IEEE Security and Privacy (S&P) conference 2026  
IEEE Security and Privacy (S&P) conference 2025  
IEEE Security and Privacy (S&P) conference 2024  
Generative AI + Law (GenLaw)'24 Workshop at ICML 2024  
Generative AI + Law (GenLaw)'23 Workshop at ICML 2023

#### **Area Chair**

Neural Information Processing Systems (NeurIPS) 2025  
Internal Conference on Machine Learning (ICML) 2025  
Neural Information Processing Systems (NeurIPS) 2024

#### **Session Chair**

DL: Robustness at International Conference on Machine Learning (ICML) 2022

#### **Reviewer**



|                                                                                   |           |
|-----------------------------------------------------------------------------------|-----------|
| International Conference on Machine Learning (ICML)                               | 2024      |
| International Conference on Learning Representations (ICLR)                       | 2024      |
| Google Research Scholar                                                           | 2023-2024 |
| Nature Machine Intelligence Journal                                               | 2023      |
| Neural Information Processing Systems (NeurIPS) + <b>Top Reviewer</b>             | 2023      |
| International Conference on Machine Learning (ICML)                               | 2023      |
| Neural Information Processing Systems (NeurIPS)                                   | 2022      |
| Nature Machine Intelligence Journal                                               | 2022      |
| International Conference on Machine Learning (ICML) + <b>Outstanding Reviewer</b> | 2022      |
| IEEE Transactions on Emerging Topics in Computing                                 | 2022      |
| Machine Learning for the Developing World (ML4D) workshop at NeurIPS              | 2021      |
| Journal of Machine Learning Research                                              | 2021      |
| Machine Learning for the Developing World (ML4D) workshop at NeurIPS              | 2020      |

### External Reviewer

|                                                     |      |
|-----------------------------------------------------|------|
| USENIX Security Symposium                           | 2022 |
| IEEE Symposium on Security and Privacy              | 2022 |
| International Conference on Machine Learning (ICML) | 2021 |
| USENIX Security Symposium                           | 2021 |
| IEEE Symposium on Security and Privacy              | 2021 |

### Mentorship & Student Researchers

|                                                |                                |
|------------------------------------------------|--------------------------------|
| <b>Ken Ziyu Liu</b><br>Stanford University     | 2024<br>PhD Student Researcher |
| <b>Saminul Haque</b><br>Stanford University    | 2024<br>PhD Student Researcher |
| <b>Enayat Ullah</b><br>John Hopkins University | 2023<br>PhD Student Researcher |

### Education

|                                                                                                                                                                                                                       |                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Bachelor of Applied Science in Engineering Science</b><br>Major in Robotics Engineering<br>Thesis: Label-Only Membership Inference Attacks as Realistic Privacy Threats<br>Graduation with Honors (cGPA 3.73/4.00) | University of Toronto<br>2015-2020 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|

### Honors and Awards

|                                                                                                                                                                |                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Schulich Leaders Full Scholarship</b><br>\$100,000 Value<br>Awarded on the basis of academic achievement and leadership to students pursuing a STEM degree. | University of Toronto<br>2015-2020 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|

**Class of 9T7 Award**

\$4000 Value

University of Toronto

2017

Awarded on the basis of academic achievement and leadership.

**Director's Summer Research Opportunities**

\$5000 Value

University of Toronto

2016

Awarded to fund a summer research opportunity in Canada at the Institute for Biomaterials and Biomedical Engineering.

**Burger King Scholarship**

\$1500 Value

University of Toronto

2015

Awarded on the basis of academic achievement and leadership.

**University of Toronto Scholarship**

\$6000 Value

University of Toronto

2015

Awarded on the basis of academic achievement.

## Competitions

---

**Undergraduate Science Case Competition (SCINAPSE)**

(Finalist of 2) of 250+ teams. Upper Year Division.

Western University

2017

**Microsoft Azure Machine Learning Case Competition**

(1st) of 20+ teams.

University of Toronto

2017

**UTEK Consulting Competition**

(Semi-Finalist) of 20+ teams.

University of Toronto

2016

**The Game, Engineering Design Competition**

(1st) of 10+ teams. \$10,000 value.

University of Toronto

Sept. 2015 - Mar. 2016

## Community Outreach

---

**Public Software**

Google Research: Main Owner of [Multi-Epoch Matrix Factorization package](#)

2023

Google Research: Owner of [Private Linear Compression](#)

2022

TensorFlow Privacy: Sole Contributor of [Bolt-On Method](#) for Differentially Private Training

2019

**CleverHans Blog**

*Arbitrating the integrity of stochastic gradient descent with proof-of-learning*

2021

*Beyond federation: collaborating in ML with confidentiality and privacy*

2021

*Teaching Machines to Unlearn*

2020

**Personal Blog**

*How to do Machine Unlearning*

2021

*Teaching Machines to Unlearn*

2020

## Community Service and Leadership

---

**University of Toronto Consulting Association, University of Toronto**  
*Director of Volunteer Consulting Group*

*University of Toronto*  
*2017-2018*

**FoodSkrapp Startup**  
*Co-Founder, CEO, and Software Developer*

*Own Incorporation*  
*2016-2017*

**You're Next Career Network**  
*Director of Business Development, Startup*

*University of Toronto*  
*2016-2017*

**Board of Directors**  
*Youth Advisor*

*Plan Canada*  
*2015-2017*

**Youth Advisory Council**  
*Member*

*Plan Canada*  
*2014-2017*

### **Technical skills**

---

|                          |                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Proficient in:</b>    | Python, C                                                                                                                    |
| <b>Familiar with:</b>    | Java, MATLAB, Perl, SQL, Elasticsearch, JavaScript                                                                           |
| <b>Python libraries:</b> | TensorFlow, Jax, Pax, SeqIO, T5X, PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow Federated, TensorFlow Privacy |

### **Soft skills**

---

|                      |                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Communication</b> | I focus on communicating complex ideas in a way anyone can understand.                                                           |
| <b>Teamwork</b>      | I care about being considerate and sharing responsibility in effective ways. Evidenced by 12 peer bonuses and 2 kudos at Google. |
| <b>Leadership</b>    | I believe that identifying strengths and clearing runways enables success.                                                       |